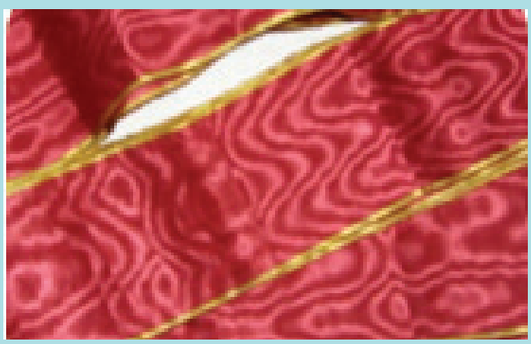
- To close the open threads
- To decrease the air permeability
- To increase the fabric clarity
- To flatten the slubs
- To modify the fabric surface by embossing

What do you mean by Moiré?

Moiré, is a textile material with a wavy (watered) appearance. It is produced mostly from silk, and sometimes from wool, cotton and rayon. The characteristic watery appearance is achieved by calendaring.



In general a calendaring machine has 2 to 7 rollers with most common being the 3 bowl rollers. Less number of bowls is used for lightweight fabrics whereas more numbers of bowls are used for calendaring heavy weight fabrics. The bowls are made with alternating hard steel and elastic. The elastic bowl is usually made from compressed paper or compressed cotton, however a lot of modern calendaring machines are made with Nylon 6 covering. This provision is given so that there remain resiliency between the two consecutive rollers and the compression remains uniform. Heating arrangements are made via steam circulation chamber.



Figure 1.2 Three Roll Calendaring Machine

The process parameters that can be controlled during calendaring are speed of the fabric, speed of the rollers and the surface of rollers. Different types of calendared effects are: surface glazing, ciré effect, moiré effect, Schreiner effect and embossing effect.

Different treatments are given to the fabric before calendaring. For example, to obtain a glazed finish like chintz (polished cotton), the fabric is first saturated in starch or wax or resin solution and is dried before calendaring. Starch or wax gives a temporary finish to the fabric but if the fabric is saturated in resin, it gives a durable glaze.

Calendared fabrics with their glossy or wet look are produced in the same manner as glazing. The fabrics are coated with a wax or resin before being calendared with heated rollers. When thermoplastic fibres are used, the fibre surface that comes in contact with the metal rollers melts and flattens slightly and produces a highly polished fabric. Examples of calendared fabric are taffeta, satin or tricot, silk or silk blend fabrics.

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